...ma Aslam Sheet No. 18456

Roll No. L15-4417 Signature

Section EE-2A Date

Q#1. $y[n] = x[n] * h[n]$

$x[n] = \left(\frac{1}{2}\right)^n u[n] \qquad h[n] = u[n]$

$\frac{100}{100}$

$$y[n] = \sum_{k=-\infty}^{\infty} x(k)\, u(n-k)$$

$$= \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k u[k]\, u[n-k]$$

$$= \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k u[n-k]$$

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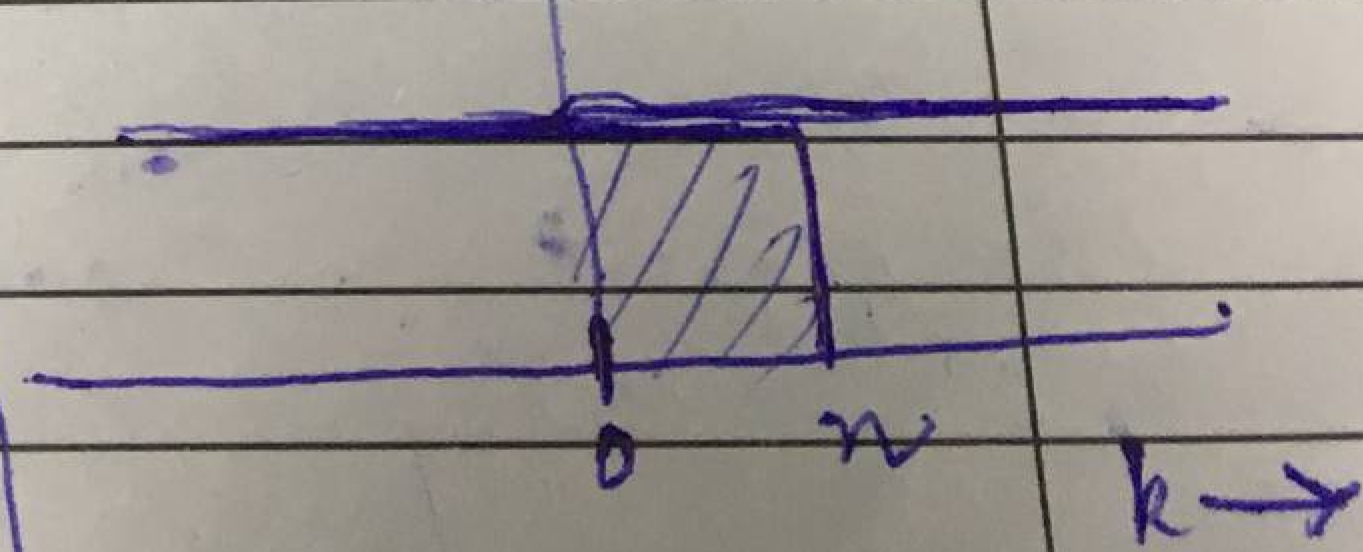
$$= \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k = \frac{1-\left(\frac{1}{2}\right)^{n+1}}{1-\frac{1}{2}}$$

$$y[n] = \begin{cases} 0, & n < 0 \\ \dfrac{1-\left(\frac{1}{2}\right)^{n+1}}{\frac{1}{2}}, & n > 0 \end{cases}$$



EE 302: Digital Signal Processing
Midterm 1

February 23, 2018

Problems

1. Compute the closed-form expression for $y[n]$:

$$y[n] = x[n] * h[n],$$

$x[n-l]$

where

$$x[n] = \left(\frac{1}{2}\right)^n u[n], \quad h[n] = u[n].$$

2. Determine the range of values of b for which the LTI system

$$h[n] = b^n u[n]$$

$b<1$

is stable. Formula?

$(-2)^{-n} u[-n]$

3. Compute the autocorrelation function $r_{xx}[l]$ of

$$x[n] = 2\delta[n] + \delta[n-1] + \delta[n-2] + 2\delta[n-3].$$

4. Compute the autocorrelation function $r_{xx}[l]$ (in closed form) of the energy signal

$$x[n] = \left(\frac{1}{2}\right)^n u[n].$$